Sama Sayed Zain

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SUMMARY

Software Engineering student at Helwan University (GPA 3.5/4.0, expected May 2027) with practical experience in full-stack and backend development using Node.js, Express.js, NestJS, React.js, and MongoDB. Completed 100+ hours of backend training with top honors (98% score). Proven track record in API design, database architecture, team collaboration, and technical mentoring.

TECHNICAL SKILLS

- **Backend Development:** Node.js, Express.js, NestJS, RESTful APIs, JWT Authentication, bcrypt, Microservices, Middleware, ASP.NET Core
- **Databases & ORM:** MongoDB, Mongoose, MySQL, Sequelize, SQL Server, Entity Framework Core
- **Frontend Development:** React.js, HTML5, CSS3, JavaScript (ES6+), TypeScript, Tailwind CSS, Bootstrap, Angular
- **Languages & Core CS:** TypeScript, JavaScript, C++, C#, Java, C, Data Structures, Algorithms, OOP, Design Patterns
- **Tools & Practices:** Git, GitHub, Postman, VS Code, Webpack, RxJS, PySpark, Agile Methodology, CI/CD

EXPERIENCE & TRAINING

Backend Development Trainee | *Route Academy, Cairo* 2025

- Mastered Node.js, Express.js, NestJS, and MongoDB across 3+ production-ready projects utilizing MVC architecture and clean code principles.
- Engineered 5+ RESTful API endpoints featuring JWT authentication, bcrypt hashing, custom middleware, and schema validation; reduced auth error rate by 40%.
- Implemented automated email verification and secure password reset workflows, cutting manual account recovery steps by 3.
- Documented and tested 10+ API collections using Postman, achieving 100% endpoint coverage with structured error handling.

MEAN Stack Web Development Trainee | *National Telecommunication Institute (NTI)* Summer 2025

- Graduated with a 98% final score — top of cohort — after constructing 2 full-stack applications with MongoDB, Express.js, Angular, and Node.js.
- Developed authenticated RESTful APIs with full CRUD operations, integrating 4+ modules within a Git/GitHub collaborative workflow.
- Gained core competencies in technical scoping, deliverable management, and freelancing best practices.

Frontend Committee Member & Mentor | *IEEE FCIH Student Branch, Cairo* 2024 – 2025

- Mentored 20+ participants at TechVerse Bootcamp 3.0 (Sep 2025) across 14 days, delivering 5+ structured sessions in JavaScript and React.js.
- Awarded Certificate of Appreciation and recognized as one of the Best Members of IEEE FCIH for Season 25.

PROJECTS

Sara7a App — Anonymous Messaging REST API github.com/Sama-Zain/Sara7a_App

- Architected a secure Node.js/Express REST API supporting end-to-end encrypted anonymous messaging.
- Implemented bcrypt hashing and AES-256-CBC encryption for user data protection alongside optimized Mongoose schemas.

Social Media Platform Backend API github.com/Sama-Zain/Social-Media

- Engineered a scalable TypeScript/Express/MongoDB backend handling user sessions, authentication, and database modeling.

E-Commerce Platform Backend github.com/Sama-Zain/E-commerce

- Built a modular NestJS/TypeScript backend integrating Socket.IO for real-time order processing and instant notification pipelines.

SkillBridge — Freelance Marketplace skill-bridge-phi-one.vercel.app

- Collaborated with 2-person team on a full-stack platform (React/Vite frontend, .NET Core backend), shipping provider dashboard managing 3+ order lifecycle stages.

EDUCATION

B.Sc. Computer Science & AI — Software Engineering | *Helwan University (FCIH)* Expected May 2027

Key Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Software Architecture.

CERTIFICATIONS & KEY ACHIEVEMENTS

- Route Academy Backend Development Certificate (2025) — Node.js, Express.js, NestJS, MongoDB, JWT.
- NTI MEAN Stack Web Development Certification (Summer 2025) — Final Score: 98%.
- ITIDA & EYouth Freelance Training Program — Client Management & Deliverable Scoping.
- IEEE FCIH — Certificate of Excellence & Best Member Season 25.
- Commercial International Bank (CIB) Virtual Summer Internship Program (July 2026).

PROFESSIONAL SKILLS & LANGUAGES

Soft Skills: Problem Solving, Analytical Thinking, Team Leadership, Adaptability, Time Management, Conflict Resolution.
Languages: Arabic (Native), English (Professional Working Proficiency).